Andrew Hua

412-918-6045 | ajhua@andrew.cmu.edu | ajhua.dev
linkedin.com/in/andrew-j-hua | github.com/ajhua-dotcom

Education

Carnegie Mellon University Pittsburgh, PA
B.S. in Computational and Applied Mathematics, B.S. in Artificial Intelligence Aug 2024 – May 2028

- Cumulative GPA: 3.8/4.0; Dean's List, High Honors (all semesters)
- Awards: 2nd Place, Harvard-MIT Mathematics Tournament; Champion, Canadian Intermediate Mathematics Contest (top score of 9,000+ worldwide); 4x AIME Qualifier
- Selected Coursework: Foundations of Software Engineering, Computer Systems, Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Autonomous Agents, Theoretical Computation

Experience

S&P Global New York, NY
Software Engineer Intern Jun 2026 – Aug 2026

- Developed MCP servers and a conversational agent from scratch (Python, internal LLM) that let the CTO and senior leadership query 3M+ error logs, latency events, and tickets directly, cutting turnaround from days/weeks of manual Databricks/Splunk analysis to seconds.
- Re-architected the agent into a multi-agent workflow (Python, AWS Strands Agents) that automates reliability analysis and visualization for credit ratings applications serving 2M+ users and 40k+ employees; used by leadership to direct SRE and PM team priorities.
- Built an LLM-based static analyzer (Python, RAG) that reviews database queries against 600+ pages of Oracle, Postgres, and MongoDB best practices, flagging unsafe or inefficient patterns in CI via an internal dev platform.

AnthrAI San Francisco, CA
Software Engineer Intern (Part-time, Remote) Dec 2025 – May 2026

- Built web-navigation agents (Playwright, Gemini) that parse the DOM/accessibility tree and plan multi-step actions, simulating 6 configurable user personas with 90% action-sequence similarity to real users.
- Architected the testing platform's orchestration layer (Node.js, Redis, Docker), running 50+ concurrent headless-browser agents with retries and per-run isolation to execute 12k+ agent runs across 800+ A/B and hypothesis-test batches.
- Developed an agent evaluation harness (Python, SQL) of synthetic websites with varied interactive components and layouts to test navigation reasoning; traced failures to missed components and action loops, then added state recovery and loop detection, cutting failures 25% and agent steps/API cost 40%.

Projects

Dynamic Memory Allocator in C Mar 2026

- Designed and implemented a custom dynamic memory allocator in C (malloc/free/realloc/calloc), evolving the free-block data structure from a linear implicit list into a segregated free list indexed by size class, increasing throughput 65x and memory utilization to 74% via block coalescing and footer elimination.
- Authored a heap consistency checker to validate block alignment, header integrity, and free-list correctness, enabling systematic detection of memory corruption.

RNN Language Model with Self-Attention Nov 2025

- Built an RNN language model with self-attention, a custom tokenizer, and greedy/stochastic decoding, trained on 100K short-story examples (10K held out) across 20 configurations on cloud GPUs.
- Achieved a final validation loss of 2.30 (≈10 perplexity); tracked and visualized evaluation metrics with NumPy and Matplotlib.

Technical Skills

Languages & Frameworks: Python, C, C++, Java, SQL, TypeScript, Node.js, Next.js, React, Angular, Playwright
AI/ML: PyTorch, NumPy, Pandas, Matplotlib, LLMs (Gemini), RAG, MCP, AWS Strands Agents
Data & Infra: Databricks, Splunk, Postgres, MongoDB, Redis, Docker, CI/CD, Git, Linux